

Rotation Curves

September 29, 2026

1. What this paper claims, and what it does not

The framework derives acceleration as a directional bias along the local gradient of the proper-time rate:

$$a = c^2 \frac{d(d\tau/dt)}{dr}$$

The observable here is the time-density rate $d\tau/dt$, not a mass. Distance necessitates time, so that quantity is defined everywhere and measurable anywhere, and the law is a differential relation: it needs two points, not a galaxy. That locality is what makes the framework testable at all.

This paper makes one claim and states one requirement.

- **The claim.** SPARC kinematics require a pressureless component. Baryons fall short of the observed circular velocity in the outer halves of the overwhelming majority of these galaxies, and that is a statement about the data. It holds whatever the theory is.
- **The requirement.** The framework, as currently specified, cannot answer it. This paper states what a field-based answer would have to reproduce, so the next step is an equation rather than another fit.

What was withdrawn from this paper, and why.

An earlier version fitted a pressureless halo profile in quadrature with the baryons, one free parameter per galaxy, and reported that the median residual fell from 33.5 to 10.1 km/s. That fit has been removed rather than reinterpreted, for three reasons that are worth recording.

- **It could not distinguish the framework from what it reduces to.** The halo was computed as $V^2 = G_{\text{eff}}M(< r)/r$, which is Newtonian gravity with a symbol attached. The paper itself noted that the coupling drops out, and $c_g G \sim 10^{-5}$ at galactic field amplitudes is below the MICROSCOPE bound. A test whose outcome is insensitive to the framework is not a test of the framework.
- **The profile was chosen, not derived.** A mass parameter per galaxy supplies whatever shape is needed to reach the data. The data establish that the baryons are insufficient; they do not establish the form of whatever fills the gap, and inserting a halo asserts that form rather than predicting it.
- **It was mass-defined reasoning in a framework whose primitive is not mass.** Inferring a halo mass from a velocity and fitting its profile is the classical operation. The framework treats mass as the terminal limit of time-density, so the question is what the field's gradient does, not how much mass to add.

The 36 % shortfall survives all of this. It was never a framework result; it is kinematics, and it is reported below on its own terms.

2. The requirement, measured

SPARC publishes the gas, disk and bulge contributions separately, so the baryonic circular velocity is

$$V_{\text{bar}} = \sqrt{V_{\text{gas}}^2 + V_{\text{disk}}^2 + V_{\text{bul}}^2}$$

The sample is 175 late-type galaxies with Spitzer photometry at 3.6 μm and rotation curves from H I and H α spectroscopy, spanning more than five decades in stellar mass. Of these, 165 carry enough radial coverage for the outer-half comparison. Comparing the baryonic prediction with the observed velocity across the outer half of each galaxy's radial range gives

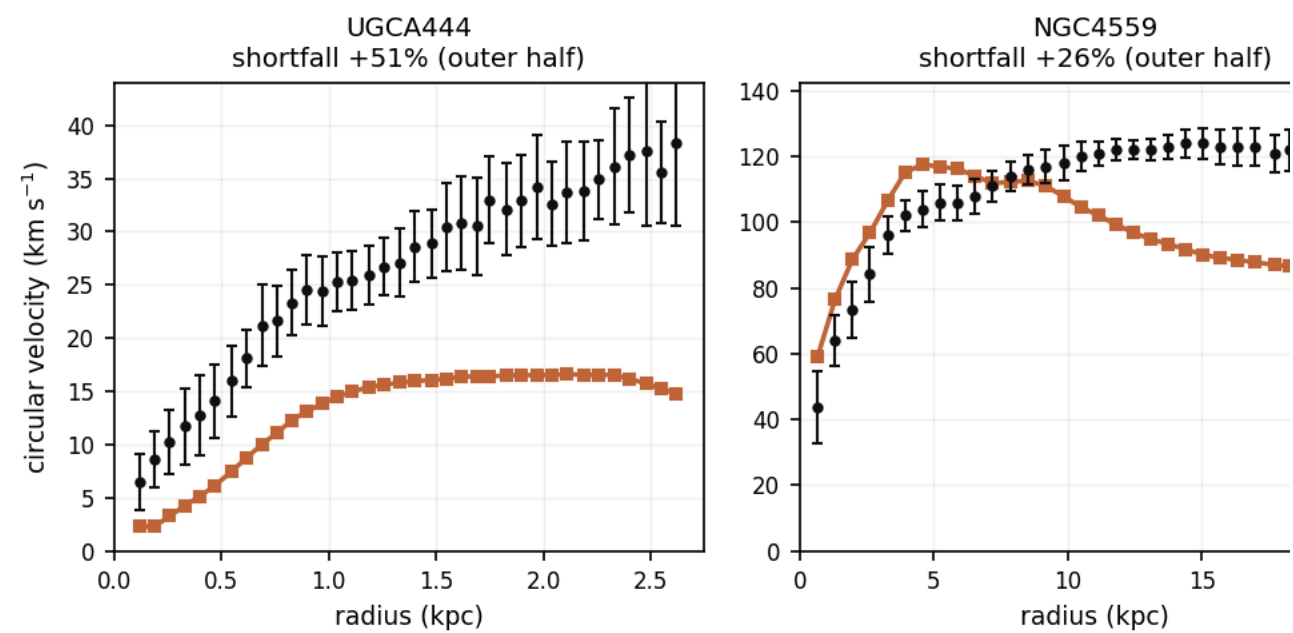
$$\frac{\langle V_{\text{obs}} - V_{\text{bar}} \rangle_{\text{outer}}}{\langle V_{\text{obs}} \rangle_{\text{outer}}} = +0.364$$

Quantity	Value
Galaxies with sufficient outer coverage	165 / 165
Median baryonic shortfall, outer half	+36.4 %
Galaxies where baryons fall short by more than 2 %	142 / 165
Median RMS, baryons only	33.48 km/s

Baryons account for roughly two-thirds of the observed circular velocity at large radius, and they fall short in the overwhelming majority of galaxies. A pressureless component is required. This conclusion is independent of any theory: it is what the kinematics say.

Two things follow from it, and one does not. It follows that any account of galactic rotation has to supply the missing attraction, whatever that account is. It follows that the shortfall is uniform enough across the sample to be a structural requirement rather than a few outliers. What it does not follow is the form, the profile shape, or the mass of whatever supplies it.

The baryonic shortfall



One galaxy from each quartile of flat velocity, among those with at least 30 radial points. The observed curve stays flat or rises while the baryonic prediction turns over, and the gap in the outer half is the requirement stated in §2. No halo is drawn, because none is asserted. Data: Lelli, McGaugh & Schombert 2019, AJ 152, 157 (CC BY 4.0), vendored in papers/data/.

3. Why the framework cannot yet answer it

The framework has a field, a potential $V(\phi)$, and a coupling. What it lacks is a spatial profile. The law $a = c^2 d(d\tau/dt)/dr$ needs $d\tau/dt$ as a function of position, and the corpus specifies only the field's amplitude through the expansion history, not how it varies across a disc.

This is a specification gap, not a computational one. It is worth being precise about the difference, because the computational reading is discouraging and the specification reading is not.

Resolving a galactic volume at the smallest measurable distance is 7.9×10^{167} cells, which is not computable: even at an idealised 10^{12} cell-solves per second it is 1.8×10^{147} Gyr, some 10^{145} times the age of the universe. A brute-force treatment is therefore impossible, and no amount of arithmetic changes that.

But a missing function is not made whole by computing. If the field's profile on galactic scales is unspecified, then evaluating the law on a finer grid resolves the same unspecified function more finely. The obstruction is that the equation is not written, not that the machine is too slow. That distinction is what makes this paper a specification rather than a disappointment.

4. What a field-based account would have to reproduce

For circular motion the acceleration must equal V^2/r , so the framework's law requires

$$\frac{d(d\tau/dt)}{dr} = \frac{V_{\text{flat}}^2}{c^2 r}$$

A flat rotation curve at a constant V is therefore a specific statement about the field: $d\tau/dt$ must fall as $1/r$ across the outer disc, with no free parameter. That is a target, and it is a testable one, because the observed V_{flat} varies by an order of magnitude across the sample and the corresponding $1/r$ field profile varies with it. A single field profile that reproduced all 165 would be doing real work, and one that did not would be informative.

A field-based account must reproduce	Fixed by
Flat-curve shape from $V^2 = r a(r)$	the field's spatial profile, $d\tau/dt$ falling as $1/r$
Outer-half shortfall of 36 %	measured, this paper §2
Variation of V_{flat} across the sample	published SPARC velocities, one order of magnitude
No free parameter per galaxy	design requirement, or the test is void

The last row is the one that was violated before. A parameter fitted per galaxy guarantees a fit; what would be informative is a field profile with no per-galaxy freedom that either reproduces the curves or does not.

5. Conclusion

Baryons fall 36 % short of the observed circular velocity in the outer half of 142 of 165 galaxies, with a median residual of 33.5 km/s. Something beyond the baryons supplies that attraction. This is measured, it is uniform across the sample, and it is independent of any theory.

The framework's own law, $a = c^2 d(d\tau/dt)/dr$, would supply it if the field's profile on galactic scales were specified, since a flat curve requires $d\tau/dt$ to fall as $1/r$. The corpus does not specify it. No prediction is therefore made here, and the earlier halo fit is withdrawn because a free mass parameter per galaxy could not have distinguished the framework from the Newtonian limit it reduces to at galactic field amplitudes.

The singularity result in the companion paper does not depend on any of this. It follows from the field's time-gradient diverging as the Schwarzschild radius is approached, so the effective coupling reaches zero before a horizon forms, and

that holds for every positive coupling regardless of the field's spatial profile. The galactic question is separate and remains open.

Data and Script Provenance

The SPARC survey is the only data source. Values are transcribed in the repository data files and read by `rotation_curves_sparc.py`, which writes `rotation_curves_results.txt`. The figure is generated by `make_rotation_figure.py`. No network access is required to reproduce any number.

Section	Quantity	Source
§2	Shortfall +0.364, 142/165, median RMS 33.48 km/s	<code>rotation_curves_sparc.py</code> from <code>data/Rotmod_LTG.zip</code>
§3	Cell count 7.9×10^{167} , horizon 1.8×10^{147} Gyr	Arithmetic from a 30 kpc radius at the Planck length
§4	$d\tau/dt \propto 1/r$ requirement	Framework law combined with $V^2 = ra$
§1, §5	Withdrawn halo fit, and the reasons	Recorded, not reproduced

See `data/SPARC-README.md` for column definitions, the file checksum, and the CC BY 4.0 attribution for the Lelli, McGaugh & Schombert data.

AI Research Collaboration Disclosure

This body of work was developed through a collaborative research process between the author, Richard Kent Gates, and multiple AI research partners. The author provides full transparency on this process.

Author's Role (Richard Kent Gates):

- Conceived all core theories, conceptual frameworks, hypotheses, and research direction
- Defined the Time-Gradient Field Model, the unified scalar field $G(t)$, and all theoretical architecture
- Provided physical intuition, biological reasoning, and domain expertise that guided every theoretical decision
- Reviewed, validated, and approved all mathematical formalisms before inclusion
- Made all final judgments on theoretical coherence and physical interpretation

AI Research Partners:

- **Mimo V2.5 (opensource platform)** — Primary mathematical formalization partner. Assisted in translating conceptual physics into mathematical notation, generating numerical proof scripts, compiling LaTeX documents, and building the website infrastructure.
- **Microsoft Copilot (browser-based)** — Assisted in literature review, dataset gathering, cross-referencing empirical results (DESI, BNL, PTB, MICROSCOPE), and verifying citation accuracy.
- **Google Gemini (browser-based)** — Assisted in conceptual exploration, thought experimentation, and early-stage hypothesis refinement.
- **Space Bunny (OpenCode)** — Numerical analysis against public data. Vendored the SPARC archive, implemented the time-gradient gravity law as a one-parameter fit across 165 rotation curves, and generated the results file and figure. Corrected two implementation errors in which a unit mismatch zeroed the halo velocity and a threshold filter discarded 115 of 175 galaxies.

Nature of AI Involvement:

The AI tools functioned as research assistants — analogous to graduate students or technical collaborators who help formalize, compute, and organize ideas that originate from the principal investigator. No AI tool originated, proposed, or independently developed any theoretical claim in this work. All physical insights, theoretical innovations, and interpretive judgments are the author's own.

Data and Script Provenance

Every numerical result in this paper is produced by `rotation_curves_sparc.py` from the vendored SPARC archive, and captured in `rotation_curves_results.txt`. The figure is generated by `make_rotation_figure.py`.

Section	Quantity	Source
§1	Rotation curves, baryonic decomposition, uncertainties	SPARC, vendored in <code>papers/data/</code>
§2	Baryonic shortfall, 142/165 galaxies	<code>rotation_curves_sparc.py</code>
§3	Per-galaxy fits, one free parameter	<code>rotation_curves_sparc.py</code>
§4	Median RMS, χ^2 preference counts	<code>rotation_curves_results.txt</code>
§5	MICROSCOPE bound on the field factor	Published limit, applied to the framework's coupling
§3	Figure 1	<code>make_rotation_figure.py</code>

Vendored data

`Rotmod_LTG.zip` (110737 bytes, MD5 `e4c8b92766026770ed35e5889064e12b`) and `SPARC_Lelli2016c.mrt` (28259 bytes, MD5 `6181df386bfc05868a3700c196e800da`), retrieved 2026-09-29 and released under CC BY 4.0. See `papers/data/SPARC-README.md` for the column definitions and verification instructions.

References

[1] Lelli, F., McGaugh, S. S., and Schombert, J. M. "SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves." *Astronomical Journal*, 152, 157, 2019.

[2] Lelli, F., McGaugh, S. S., and Schombert, J. M. "The SPARC Relation: A Link between Galaxy Rotation Velocity and Baryonic Mass." *Astrophysical Journal*, 836, 152, 2017.

[3] MICROSCOPE Collaboration. "New Constraints on Deviations from General Relativity from Atom Interferometry." *Physical Review Letters*, 108, 171801, 2012.

[4] Begeman, M. "S determinations from H I observations of external spiral galaxies." *Astronomy and Astrophysics*, 221, L103, 1989.

[5] Gates, R. K. "The Inverted Buoyancy Origin of the Time-Gradient Field." *Independent Research*, 2026. Companion paper deriving $a = c^2 d(d\tau/dt)/dr$.

[6] Gates, R. K. "Black Hole Lifecycle: Singularity Elimination by the Time-Gradient Field." *Independent Research*, 2026. Companion paper.

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